

Tuning VSC: plant transfer functions

Simplified system model

- 1 elements: related to the VSC
- 2 elements: related to the Grid
- 3 elements: related to the filter/PoC

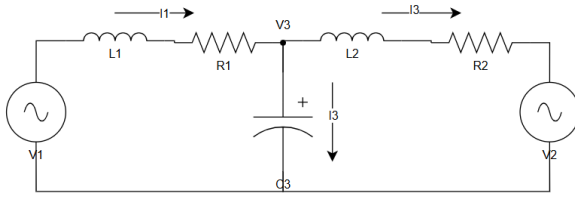


Figure 1: Basic Scheme

Analytically in abc frame represented by the following ODEs:

$$\begin{aligned}
 i_1 &= i_2 + i_3 \\
 i_3 &= C_3 \frac{dv_3}{dt} \\
 v_1 - v_3 &= L_1 \frac{di_1}{dt} + R_1 i_1 \\
 v_3 - v_2 &= L_2 \frac{di_2}{dt} + R_2 i_2
 \end{aligned}$$

The relative cascade loops for both branches are represented as follows:

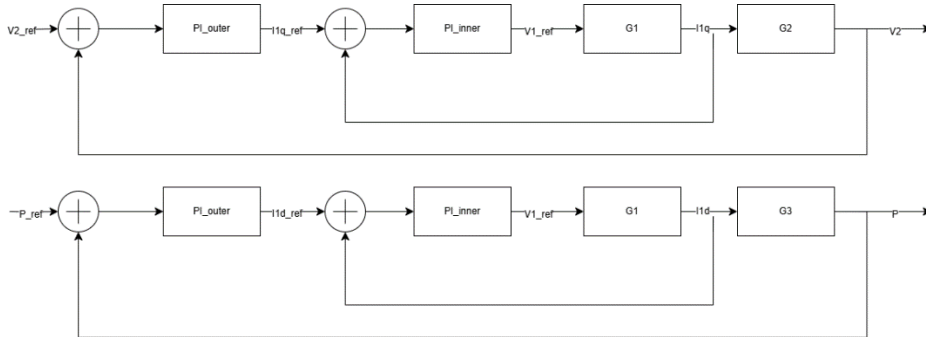


Figure 2: Control Structures

Inner loop plant (G1)

The first plant transfer function links the voltage V1 (the controlled VSC's terminal voltage) to the current I1 (the current that flows out of the converter).

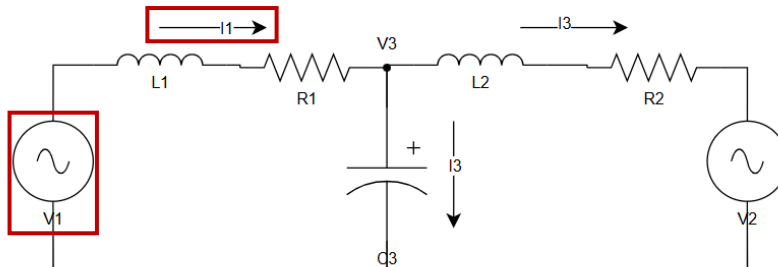


Figure 3: Scheme for Inner Loop

The relationship can be derived by hand by explicating the equations, to reduce the error, the Modified Nodal Analysis with symbolic elements (by Erik Cheever)¹. The circuit is solved algebraically by finding the currents of each branch and the voltages of each node, and then the ones of interest for the relationship are related. 📌

The scheme in Figure 2 must be represented as Netlist. 📌

Name, Starting_Node, Reaching_Node, Type/Value

- V1 1 0 Symbolic
- L1 1 2 Symbolic
- R1 2 3 Symbolic
- L2 3 4 Symbolic
- R2 4 0 Symbolic
- C3 3 0 Symbolic

Generator V2 is tuned off (replaced with a short circuit) since it is considered a disturbance.

The transfer function is :

$$G_1 = \frac{I_1}{V_1} = \frac{C_3 L_2 s^2 + C_3 R_2 s - 1}{C_3 L_1 L_2 s^3 + (C_3 L_1 R_2 + C_3 L_2 R_1) s^2 + (L_1 + L_2 + C_3 R_1 R_2) s + R_1 + R_2}$$

¹ "Analysis of Circuits - Intro," accessed June 6, 2025, <https://lpsa.swarthmore.edu/Systems/Electrical/mna/MNA1.html>.

Outer Loop that controls the PoC voltage (G2)

In this case, the output is the voltage at the PoC (V3) controlled by the converter's output current (I1).

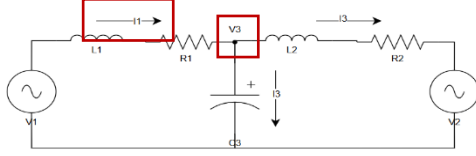


Figure 4: Scheme for Outer Loop (V)

The transfer function is:

$$\frac{V_3}{I_1} = \frac{R_2 + L_2 s}{C_3 L_2 s^2 + C_3 R_2 s + 1}$$

However, the control is done just by the I_{1q} part, and it controls just the V_{3d} component. The V_{3q} component is zero or approximately zero since the Park transform has to be aligned with the voltage at that node. Therefore, the absolute value of the voltage at the PoC is composed as follows:

$$V_{poc} = V_3 = \sqrt{v_{3d}^2 + v_{3q}^2} \cong v_{3d}$$

In order to link $i_{q1} \rightarrow v_{3d}$ the above transfer function must be converted to the dq frame, which can be done by:

$$\begin{cases} s \rightarrow (s + j\omega) \\ V_3 \rightarrow V_{3d} + jV_{3q} \\ I_1 \rightarrow I_{1d} + jI_{1q} \end{cases}$$

In particular the $V_{3q} = I_{1d} = 0$. This is verified through a simulation with fictitious gains (that keep the system stable) so the values can be visible:



Figure 5: Simplification proof

The resultant transfer function in the dq frame, where the “q” current component is linked to the “d” voltage one, is:

$$G_2 = \frac{V_{3d}}{I_{1q}} = - \frac{L_2 \omega}{C_3 L_2 s^2 + C_3 R_2 s - C_3 L_2 \omega^2 + 1}$$

Outer Loop that controls Active Power (G3)

The active power links the current output of the converter (I_1) with both V_3 and I_2 (which are used to compute the active power that flows into the grid, P).

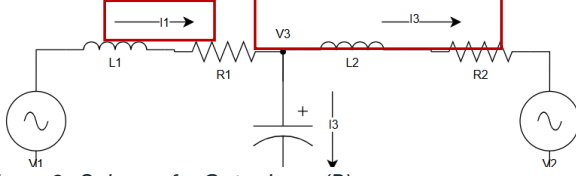


Figure 6: Scheme for Outer Loop (P)

The active power can be computed directly in the dq frame of the (3) and (2) variables and using simple equations by considering $v_{3q} \cong 0$.

$$P = v_{d3}i_{d2} + v_{q3}i_{q2} \cong v_{d3}i_{d2}$$

By linearizing the active power around the initial conditions (active power $P_0 = 0 \rightarrow I_{d20} = 0$) the relationship becomes:

$$\tilde{P} \cong P_0 + V_{d30}i_{d2} + I_{d20}v_{d3} \cong V_{d30}i_{d2}$$

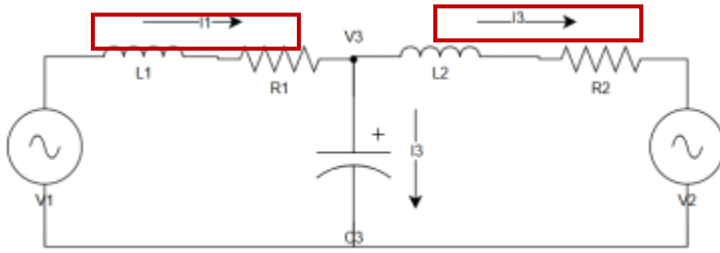
Where V_{d30} is the steady state initial voltage (most likely the nominal one).

In this case, the transfer function will be practically by relating the current I_2 and I_1 (in d frame):

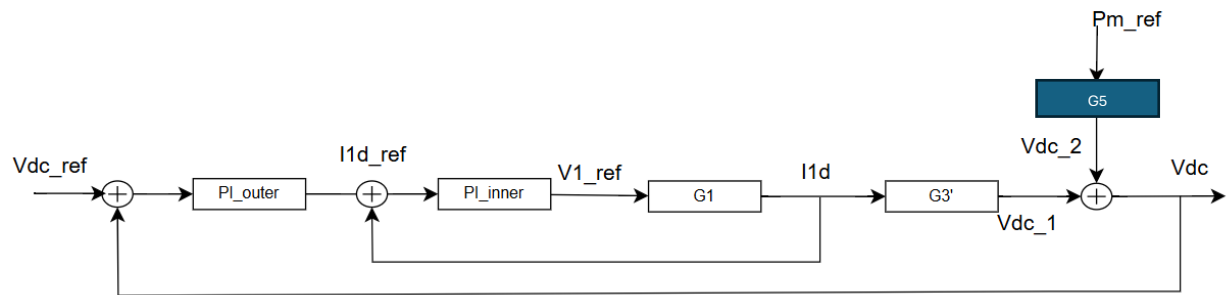
$$G_3 = \frac{\tilde{P}}{I_{1d}} = - \frac{V_{d30}}{C_3 L_2 s^2 + C_3 R_2 s - C_3 L_2 \omega^2 + 1}$$

Outer Loop controls DC side voltage controller (G3')

The DC side dynamics is an additional block that must be controlled and replaces the active power controller. Therefore, the plant G3 cannot be considered the same as before, but the active power relationship that connects the AC side variables to the linearized active power can be recycled. The connection here is $I_{1d} \rightarrow V_{dc}$.



The control loops are similar to the active power. The tuner in matlab must be set to “disturbance rejection” as objective since the V_{dc_ref} never changes



The initial DC dynamics:

$$\frac{dv_{DC}}{dt} = \frac{P_{mref}}{C_{DC}v_{DC}} - \frac{P_{ac}}{C_{DC}v_{DC}}$$

It is tuned just for the P_{ac} case because the P_{m_ref} counts as a disturbance. And the other part is linearized as follows:

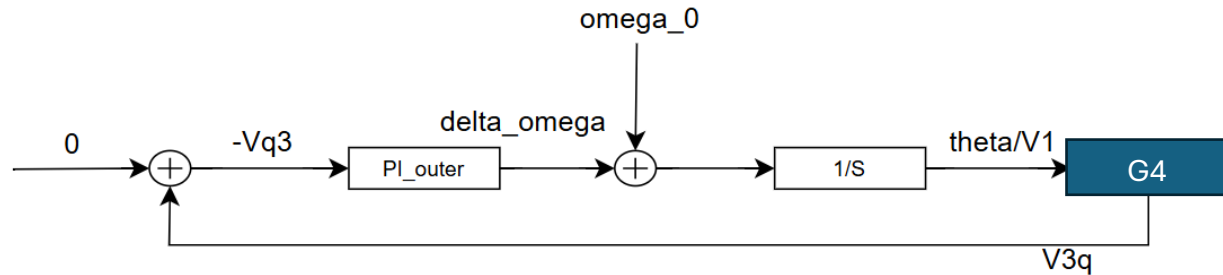
$$s\tilde{v}_{DC} = -\frac{V_{d30}i_{d2}}{C_{DC}V_{DC0}} \rightarrow \frac{\tilde{v}_{DC}}{i_{d1}} = -\frac{1}{s} \frac{V_{d30}}{C_{DC}V_{DC0}} \frac{i_{d2}}{i_{d1}}$$

Where the last part of the relationship between the currents is the same as in the active power:

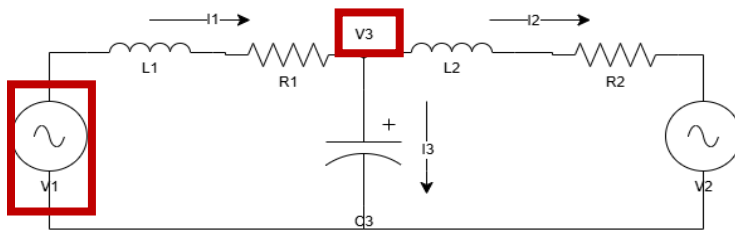
$$G'_3 = \frac{\tilde{v}_{DC}}{i_{d1}} = -\frac{V_{d30}}{C_{DC}V_{DC0}(C_3L_2s^3 + C_3R_2s^2 + (-C_3L_2\omega^2 + 1)s)}$$

PLL Tuning (G4)

The connection establishes a feedback loop between voltage V1 and voltage V3q, utilizing the q component for controller purposes. There should then be a connection between the theta applied to the controlled voltage source and the q component of the PoC voltage.



Thus, the connection should be made between these two.



In this case, since just the q part of v3 is the one of interest, it uses the same thing to convert every s into $(s+j\omega)$.

Therefore, the straight relationship becomes*:

$$G_4 = \frac{V_{3q}}{\theta} = \frac{L_2 \omega}{C_3 L_1 L_2 s^3 + (C_3 L_1 R_2 + C_3 L_2 R_1) s^2 + (-3 C_3 L_1 L_2 \omega^2 + L_1 + L_2 + C_3 R_1 R_2) s + R_1 + R_2 - C_3 L_1 R_2 \omega^2 - C_3 L_2 R_1 \omega^2}$$

- This is the most “unsure” one....